

Proper training along with professional degree would be the turnkey for a fresher. This training should be totally industry-focused. I feel an effective collaboration between the industry and academia may be fruitful for both the parties.”

—N. Chandramohan, country head-SMT division, Juki India



are the building blocks. Functionally, they are no different from leaded ones.

SMCs are mounted on the surface of the PCB. The solder joints are all the more important as these impart both electrical and mechanical connections unlike the through-hole components where the component leads sit in the plated through holes which provide certain amount of mechanical strength when soldered.

You can identify SMCs with their unique identifiers which are a combination of their dimensions (width and length) expressed in inches; e.g., 1208 component means its dimensions are 0.12 (w) x 0.08 (l) inch². Ceramic capacitors, tantalum capacitors and thick-film resistors form the core group of passive components.

There are two main categories of chip carriers: ceramic and plastic. The plastic chip carriers are primarily used in commercial applications. The ceramic packages are expensive and provide hermeticity. These are used primarily in military applications. Both the passive and active SMCs are much smaller in size than their through-hole analogues. Not only the size but also the shape of these components is modified according to the need of automated handling and better stability.

SMCs are most commonly available in flat rectangular shape. Another interesting point is the leadless format of passive components, whereas active components may have mountable lead or leadless terminals. To reduce the

size of the circuits, these components often have reduced pitches.

The current trend is towards the usage of larger pin-count packages. Packages having more than 84 pins become impractical with 50mil lead centres due to larger package area and difficulties in manufacturing. To overcome the difficulties associated with larger pin-count packages, VLSI manufacturers have adopted 25mil centre. Often, this extreme size reduction reduces the mechanical strength. Thus, it becomes a designer's challenge also.

In summary, surface-mount components serve the same functions electrically as their through-hole counterparts. However, these have smaller package area and better electrical performance. At this stage, all the components are not available in surface-mount form and therefore the designers have to choose rightly between conventional and surface-mount components.

How to mount on the surface?

The next thing you need to know is the process flow of the mounting operation. It is the simplest form of assembly. The basic steps involved are paste application, pick-and-place of components, preheat or reflow, cleaning, inspection, touchups and, last but not the least, testing and repairing related activities. Try to know the activities associated with each of these steps as well as the machineries used. Learn

how the software and hardware parts of SMT equipment are dedicated to maintain process parameters. Remember, all process parameters are focused on addressing three issues: ensure mechanical strength of the assembly, better automation without drifting of the components and higher-quality end products.

Turn to paste application. You will find various dispensing methods and adhesives available. In dispensing you will come across the terms like pin transfer, screen transfer and pressure syringe transfer. Each of these has some specific advantages and disadvantages and you have to select the most suitable according to the need of the system. For example, the pressure syringe method can handle irregular surface but requires more maintenance. Whereas, screen transfer is a comparatively simpler method but it is applicable only on flat board surfaces.

Typically, an adhesive consists of a resin and a hardener. You need to know the importance of their mixing properties. Look, no adhesive is suitable for all types of applications. So, instead of knowing them individually try to learn the process requirements and matching procedures. Here you may get acquainted with the 'pot life' or the extent of cross linking of the resin mix.

Though solders are available in many forms, solder pastes are the most suitable form for surface mounting. Most of the solder pastes used for surface mounting are lead-free in nature, leading to an environment-friendly manufacturing process.

The efficiency of soldering is dependent on the rheology of the solder paste as well as its solderability. Solderability is the property of wetting. Good wetting results in the formation of a uniform, smooth, unbroken, adherent coat of solder on the base metal. Therefore wetting is an essential prerequisite for soldering. Wetting means that molten solder diffuses into the solid surface of the metal to be soldered. It is possible only if the solder can come into immediate